

Class 11 Biology: High-Yield Diagram & Mark Breakdown

Chapter 5: Morphology of Flowering Plants (NCERT Page-wise Priority Guide)

Instructions for Students: Practice schematic representations (Aestivation/Placentation) and seed cross-sections with exact labels and corresponding plant examples.

Fig No.	Diagram / Topic	NCERT	Marks	Priority	Key Labelling Points / Concepts
Fig 5.15	Monocot Seed (Maize Grain L.S.)	Pg 67	3 M	Top Priority	Aleurone layer, Endosperm, Scutellum, Coleoptile, Plumule, Radicle, Coleorhiza
Fig 5.14	Dicot Seed (Gram Seed L.S.)	Pg 66	3 M	Top Priority	Testa, Tegmen, Hilum, Micropyle, Cotyledons, Plumule, Radicle, Embryonal axis
Fig 5.11	Types of Aestivation in Corolla	Pg 64	3 – 5 M	Top Priority	(a) Valvate, (b) Twisted, (c) Imbricate, (d) Vexillary (Standard, Wings, Keel) + Examples
Fig 5.12	Types of Placentation	Pg 65	3 – 5 M	Top Priority	(a) Marginal, (b) Axile, (c) Parietal, (d) Free central, (e) Basal + Plant examples
Fig 5.16 & 5.17	Floral Diagram & Solanaceae Formula	Pg 67, 68	3 M	Top Priority	Mother axis dot, Calyx, Corolla, Epipetalous Androecium, Bicarpellary syncarpous Gynoecium
Fig 5.3	Regions of Root-tip	Pg 59	2 – 3 M	Top Priority	Root cap, Meristematic zone, Elongation zone, Maturation zone, Root hairs
Fig 5.9	Floral Parts on Thalamus	Pg 62	3 M	High Priority	Hypogynous (superior ovary), Perigynous (half-inferior), Epigynous (inferior ovary)
Fig 5.13	Parts of a Fruit (Mango & Coconut)	Pg 66	2 – 3 M	High Priority	Epicarp, Mesocarp (fleshy in mango / fibrous in coconut), Stony Endocarp, Seed
Fig 5.4	Structure of Leaf & Venation	Pg 60	2 M	High Priority	Lamina, Petiole, Leaf base, Stipule, Reticulate vs. Parallel venation

EXAM STRATEGY & GUIDELINES

- **Direct NCERT Exercise Questions:** Ex. Q5 asks to draw labelled diagrams of Gram seed (Pg 66) & V.S. Maize seed (Pg 67); Ex. Q6 asks for Floral diagram of Solanaceae (Pg 68).
- **3–5 Mark Schematic Questions:** Aestivation (Fig 5.11) and Placentation (Fig 5.12) are frequently tested as 3-mark or 5-mark theory + diagram questions.
- **Monocot Seed Accuracy:** Never miss labelling the single shield-shaped cotyledon (*Scutellum*) and protective sheaths (*Coleoptile* & *Coleorhiza*).